Lab 8: Interrupted Time-Serier & Difference-in-Differences

PHS2000B, Spring 2026

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April 9, 2026

Overview

Interrupted Time-Series

- Introduction
- Assumptions
- Data structure & analysis

Difference-in-Differences

- Introduction
- Assumptions
- Data structure & analysis
- Recent breakthroughs in DiD

Estimands

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DID Excersize

ITS Excersize

Interrupted Time-Series

Introduction

ITS is used to evaluate a treatment that's implemented a specific point in time

- Like other quasi-experimental methods, ITS is useful for settings where conducting a RCT is not feasible.
- Requires less granular data in terms of the observational unit, making ITS more accessible using administrative or government data.
- Can be useful when evaluating a program or policy at a population-level
- Only requires data from treated-units

Introduction

How the impact is modeled should be informed by the nature of the program/shock

a) Level change at time of intervention
b) Slope change at the time of the intervention
c) Level and slope change at time of intervention
d) Lagged slope change
e) Temporary level change
d) Temporary slope change

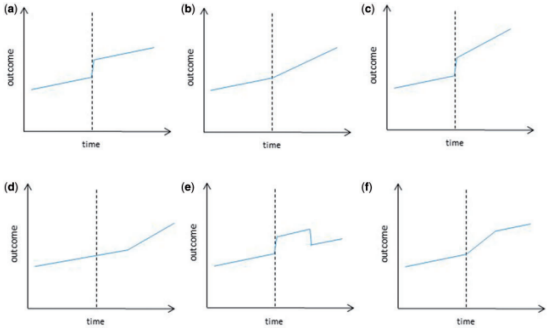
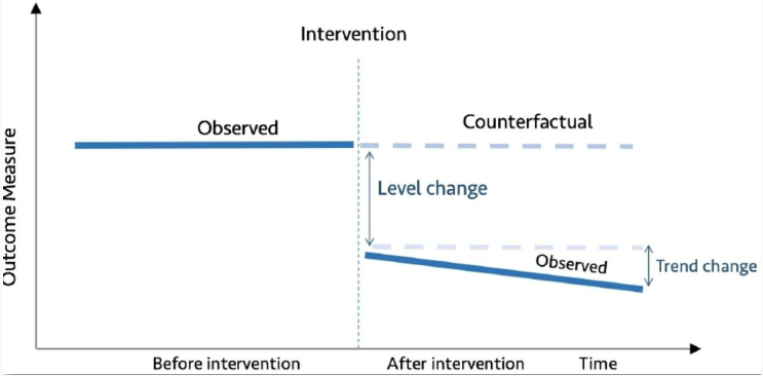


Figure 2 Examples of impact models used in ITS

(a) Level change; (b) Slope change; (c) Level and slope change; (d) Slope change following a lag; (e) Temporary level change; (f) Temporary slope change leading to a level change.

Assumptions

For standard ITS,the primary assumption is that we can use the pre-treatment trend as a counterfactual for what would have happened in the absence of treatment. Related, we have to assume treatment timing is 'as good as random'



Assumptions

$$\text{i) } Y_t(0) = \gamma_0 + \gamma_1 time_t + \gamma_2 post_t + \gamma_3 time \times post_t + v_t$$

$$\text{ii) } Y_t(1) = \alpha_0 + \alpha_1 time_t + \alpha_2 post_t + \alpha_3 time \times post_t + \epsilon_t$$

No anticipation (Pretrends would be the same regardless of intervention):

$$\alpha_0 = \gamma_0,\ \alpha_1 = \gamma_1$$

No timevarying confounding (Posttrend in the absence of intervention would be the same as pretrend):

$$\gamma_2 = 0,\ \gamma_3 = 0$$

Data structure & analysis

Data[1]

- $\geq$ 12 time points before and after intervention
- $\geq$ 100 observations per time period
- Data aggregated at the period-level, either as a count or rate

Analysis

- Center time variable at time of intervention
- Account for autocorrelation in error term
- Account for seasonality with season-fixed effects or Fourier transformation

[1] Baicker, Katherine and Svoronos, Theodore, Testing the Validity of the Single Interrupted Time Series Design (July 1, 2019). University of Chicago, Becker Friedman Institute for Economics Working Paper No. 2019-97

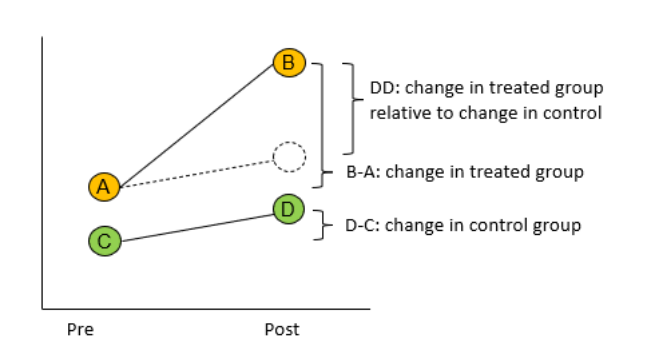
Difference-in-Differences

Introduction

- Similar to ITS, causal inference based on the timing of an intervention/shock
- If ITS' lack of a control group concerns you, DiD has got you covered! Instead of using pretrends, the **untreated units** are used to construct a counterfactual for the treated units
- One of the most popular methods for economists

Introduction

In the traditional two-period, two-unit case, the change in the untreated unit from the pre-period to the post-period is used to construct the counterfactual for the treated unit in the post-period



Assumptions

- **Parallel Trends**: trends in the the treated group would be parallel to the untreated group in the post-treatment period, had there been no intervention. **This is inherently untestable**.

- **Stable Unit Treatment Value Assumption (SUTVA)**: A unit's potential outcomes are not influenced by the treatment status of any other units' treatment status

Parallel trends

No uncommon shocks: Uncommon shocks other than the treatment of interest can violate parallel trends. If a shock is experienced by both treated and controls in the same way (common), then it will not violate parallel trends.

What are potential uncommon shocks?

Parallel ‘pre’ trends

Many economists will look at the pre-trends for treated and untreated units and assess the appropriateness of DiD based on whether these trends are parallel. However, parallel pre-trends are neither **necessary** nor **sufficient** for proveing parallel trends

- Standard statistical tests don’t allow us to ’accept’ parallel trends, can only fail to reject
- With large enough sample sizes can reject regardless of the magnitude of the deviation – but we likely care about this magnitude
- Use of equivalence tests might be more appropriate
- Flexible re-formulation of the model based on non-parallel pre-trends

SUTVA

A unit's potential outcomes are not influenced by the treatment status of any other units' treatment status. There a number of potential violations

- The intervention in the treated units causes changes in the untreated units
- Movement between treated and untreated units

Data structure and analysis

Data

- Need a minimum of two units (1 treated, 1 untreated), but more are needed for valid inference
- Need a minimum of two time periods (1 pre, 1 post), but more periods allows for comparison of pre-trends and changing effects in the post
- Data left granular/disaggregated

Data structure and analysis

- **Simple DiD model**: $Y_{it} = \alpha + \delta Treat_i + \gamma Post_t + \beta(Post \times Treat)_{it} + \epsilon_{it}$
- **TWFE, aggregate ATT**: $Y_{igt} = \delta_0 + \sum_{g=2}^{M} \alpha_g + \sum_{t=2}^{K} \gamma_t + \beta(I_{post} \times I_{Treated})_{igt} + \epsilon_{igt}$
- **TWFE, period-specific ATTs**:
  $Y_{igt} = \delta_0 + \sum_{g=2}^{M} \alpha_g + \sum_{t=2}^{K} \gamma_t + \sum_{t=-T}^{-2} \phi_t * D_{igt} + \sum_{t=0}^{T} \beta_t * D_{igt} + \epsilon_{igt}$

Which of these are the main parameters of interest?

Recent breakthroughs in DiD: Treatment heterogeneity

- Traditional TWFE estimation methods assume that there is no effect modification between real time (i.e. year) and treatment
- Violations to this assumption when treatment adoption is staggered can lead to negative weights, which can induce bias into your causal estimates
- There has been a recent boom in novel DiD estimators that are robust to this treatment heterogeneity when treatment is staggered

Recent breakthroughs in DiD: Covariate adjustments

- It is incredibly common for researchers to include covariates in the model either because of 'improved precision' or to obtain conditional parallel trends
- Econometricians have shown that one should basically never include time-group varying covariates
- There might be a need to include pre-treatment covariates to achieve conditional parallel trends when unconditional parallel trends aren't likely. However, these covariates should be included in a saturated manner or a novel estimator robust to covariate inclusion should be used.

Estimand

Estimand

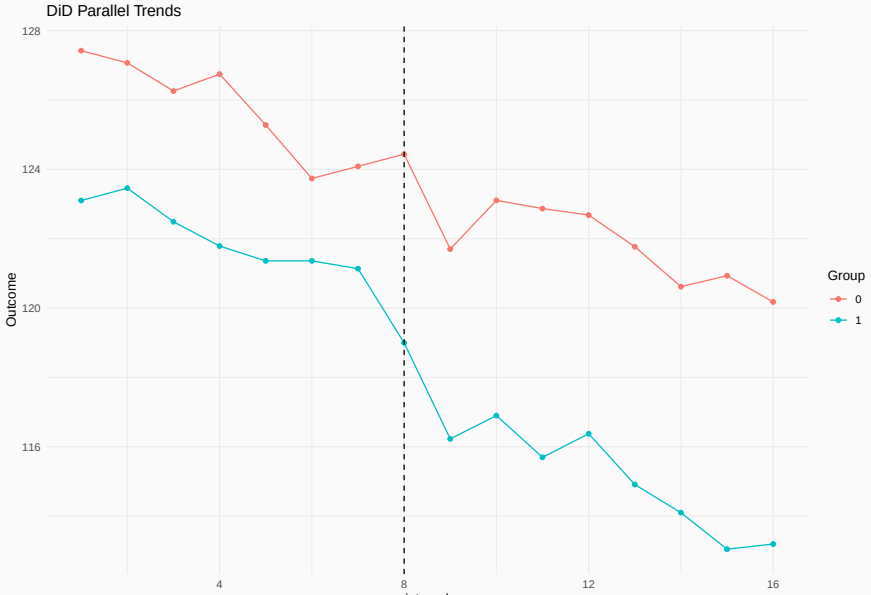
Both ITS[2] and DiD estimate the the **average treatement effect on the treated (ATT)**

$$\text{ATT} = \text{E}[Y^1 - Y^0 | T = 1]$$

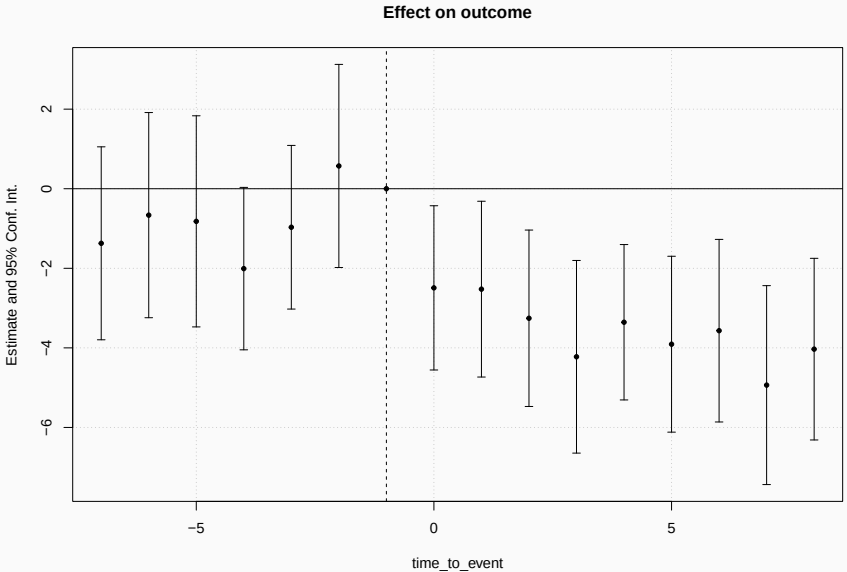
[2] Backing out the ATT from the ITS is a bit tricky. Assuming the assumptions hold, there should be a weighting scheme to estimate the ATT in the post-period.

Event study practice

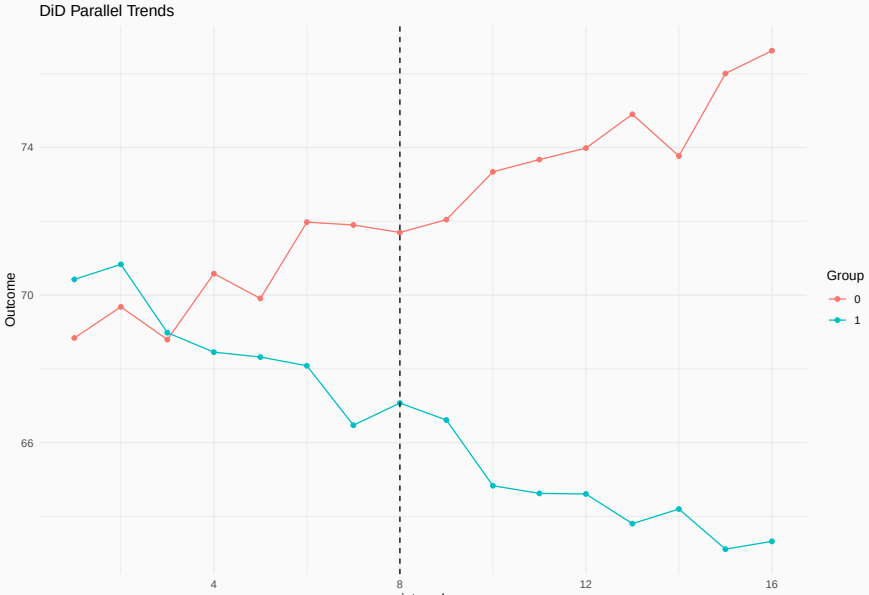
Event study practice #1



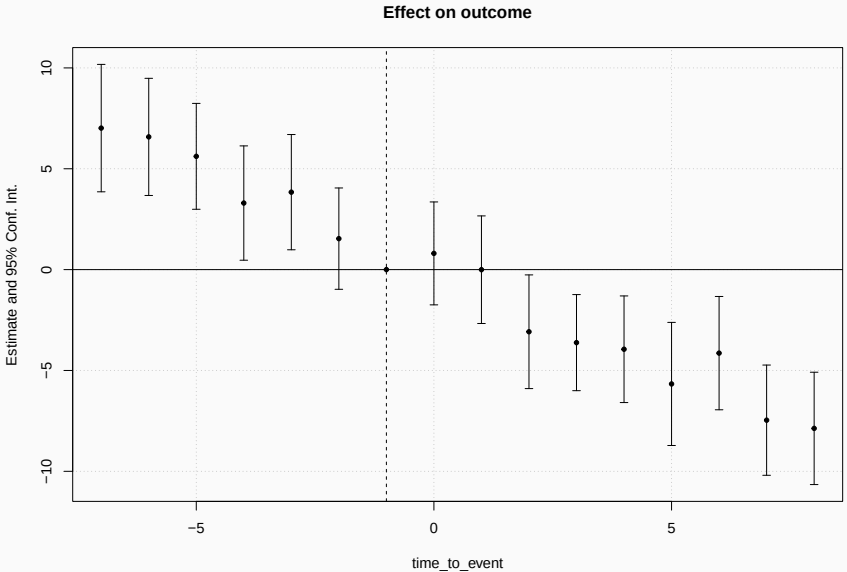
Event study practice #1



Event study practice #2



Event study practice #2



DiD Excersize

DiD Excersize

Physician quality is an important factor in the health production function. With this in mind, hospitals have incentives to acquired higher skilled personnel. We will evaluate a hypothetical intervention where half of the hospitals in your national health system implement at the same time a hiring reform where large numbers of higher skilled, more experienced physicians and nurses were rapidly hired. To explore whether this reform was effective, we look at the effect on hospital acquired infections (HAIs) as a measure of healthworker quality.

To Note: The decision for whether a hospital would undergo this hiring reform was made at the city-level.

DiD Excersize

```r
data <- read.csv("PHS2000B_lab_data.csv")
head(data)
```

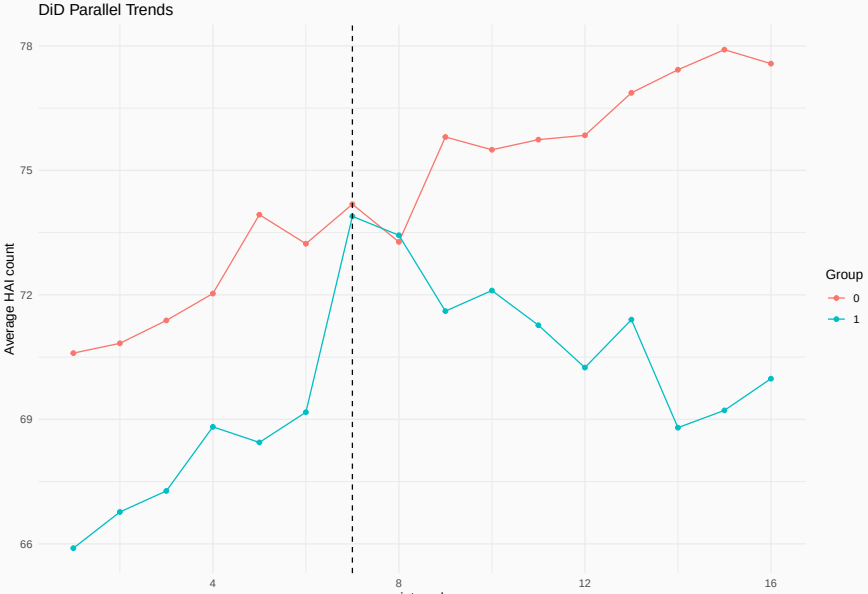
```
##   X treat hospital interval city post post_treat time_to_event city_shock
## 1 1     0        1        1    1    0          0             0         29
## 2 2     0        1        2    1    0          0             0         29
## 3 3     0        1        3    1    0          0             0         29
## 4 4     0        1        4    1    0          0             0         29
## 5 5     0        1        5    1    0          0             0         29
## 6 6     0        1        6    1    0          0             0         29
##   hai_count
## 1        61
## 2        78
## 3        73
## 4        82
## 5        54
## 6        75
```

DiD Excersize

```
library(dplyr)
library(ggplot2)
# 1. Calculate average outcome by group and year
plot_data <- data %>%
  group_by(interval, treat) %>%
  summarize(hai_count = mean(hai_count, na.rm = TRUE))

# 2. Create the parallel trends plot
plot<-ggplot(plot_data, aes(x = interval, y = hai_count, color = factor(treat))) +
  geom_line() +
  geom_point() +
  geom_vline(xintercept = 7, linetype = "dashed") + # Vertical line at treatment
  labs(title = "DiD Parallel Trends", y = "Average HAI count", color = "Group") +
  theme_minimal()
```

Data Excerize



Data Excerize

How would you interpret the estimate of interest?

```r
library(fixest)
ATT<-feols(hai_count ~ post_treat | city + interval, cluster = ~hospital, data = data)
ATT
```

```
## OLS estimation, Dep. Var.: hai_count
## Observations: 8,000
## Fixed-effects: city: 50,  interval: 16
## Standard-errors: Clustered (hospital)
##            Estimate Std. Error  t value Pr(>|t|)
## post_treat   -0.544   0.445094 -1.22221   0.2222
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 9.81834     Adj. R2: 0.45637
##                 Within R2: 1.798e-4
```

Data Excerize

The hiring reform reduced the average number of HAIs in the post-period by 0.5 cases in those hospitals that implemented it. The effect is not statisticaly significant at an alpha threshold of 0.05

```
library(fixest)
ATT<-feols(hai_count ~ post_treat | city + interval, cluster = ~hospital, data = data)
ATT
```

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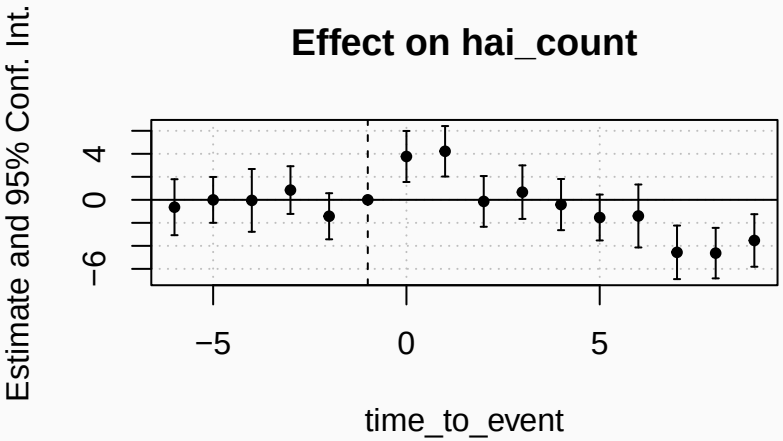
## RMSE: 9.81834     Adj. R2: 0.45637
##                 Within R2: 1.798e-4
```

Data Excerize

Based on the graph illustrating the trends of the two groups, does this make sense?

Data Excerize

```
eventstudy <- feols(hai_count ~ i(time_to_event, treat, ref = -1) | hospital + interval, cluster =~city, data = dat
iplot(eventstudy)
```



Data Excerize

How would we interpret this event study plot?

Data Excerize

How do we feel about the parallel trends assumptions?

Thank you everyone for putting up with me (and thank you Austin Tucker for letting me rip so much from his slides)

Extra enrichment

Here are two fun resources for folks who want to dive just a little deeper:

- Comparison of CITS and DiD: https://pmc.ncbi.nlm.nih.gov/articles/PMC8522572/
- Comparison of fixed-effects and random effects: https://libguides.princeton.edu/R-Panel
- Overview of current DiD methods in health research: https://pubmed.ncbi.nlm.nih.gov/41055548/

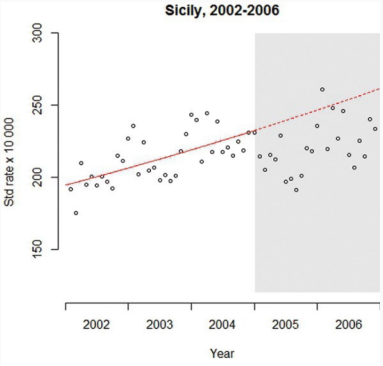
ITS Excersize

ITS excersize

Let's revisit the example of the effect of a smoking ban on acute coronary events (ACEs)

Lopez Bernal et al (2017) provide data from Barone-Adesi et al (2011) used to estimate the effects of an Italian smoking ban in 2005 on hospital admission rates for acute coronary events (ACEs). Included data are ACEs in Sicily from 2002-2006 among ages 0-69 years

With 4 years of data we have 59 observed months of ACEs



ITS excersize

```r
data <- read.csv("sicily.csv")
head(data)
```

```
##   year month aces time smokban      pop   stdpop
## 1 2002     1  728    1       0 364277.4 379875.3
## 2 2002     2  659    2       0 364277.4 376495.5
## 3 2002     3  791    3       0 364277.4 377040.8
## 4 2002     4  734    4       0 364277.4 377116.4
## 5 2002     5  757    5       0 364277.4 377383.4
## 6 2002     6  726    6       0 364277.4 374113.1
```

```r
# Smoking ban begins in time=37
table(data$smokban, data$time)
```

```
##
##     1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28
##   0 1 1 1 1 1 1 1 1 1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
##   1 0 0 0 0 0 0 0 0 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
##
##    29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53
##   0  1  1  1  1  1  1  1  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
##   1  0  0  0  0  0  0  0  0  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
##
##    54 55 56 57 58 59
##   0  0  0  0  0  0  0
##   1  1  1  1  1  1  1
```

ITS excersize

Let's say we hypothesize that there would only be level shift in ACEs due to policy implementation, bit no change in trend. As such, we would estimate the following model:

$$ACE_rate_t = \alpha + \beta time_centered_t + \delta smoke_ban_t + \epsilon_t$$

ITS excersize

```r
# Lopez Bernal compute the standardized rates for their analysis
# This is not fundamental to ITS, but we apply their code:
data$rate <- with(data, aces/stdpop*10^5)
data$time_centered <- data$time - 37

model_1 <- lm(rate ~ time_centered + smokban, data=data)
summary(model_1)
```

```
##
## Call:
## lm(formula = rate ~ time_centered + smokban, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -25.327 -10.825  -3.043   9.840  39.142
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   232.9215     4.4156  52.749  < 2e-16 ***
## time_centered   1.0621     0.2019   5.261 2.34e-06 ***
## smokban       -24.0933     7.0500  -3.417  0.00118 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.13 on 56 degrees of freedom
## Multiple R-squared:  0.359,  Adjusted R-squared:  0.3361
## F-statistic: 15.68 on 2 and 56 DF,  p-value: 3.916e-06
```

How would you interpret these coefficients?

ITS excersize

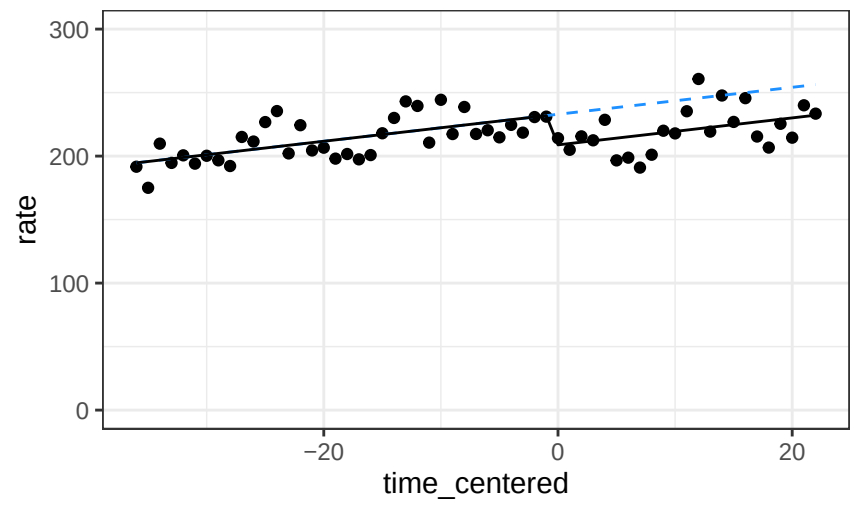
$$ACE_rate_t = \alpha + \beta time_centered_t + \delta smoke_ban_t + \epsilon_t$$

- $\boldsymbol{\alpha}$: The average rate of ACEs per 10,000 people right before policy implementation is 233
- $\boldsymbol{\beta}$: The average monthly rate of change in the average rate of ACEs per 10,000 people is 1
- $\boldsymbol{\delta}$: Policy implementation immediately reduces the level of ACEs per 10,000 people by 24.

ITS excersize

```r
library(ggplot2)

# We generate predicted values based on the model in order to create a plot
pred1 <- predict(model_1, type="response", data)

# Generate new dataset to estimate the counterfactual

datanew <- data
datanew$smokban <- 0
pred1b <- predict(model_1, type="response", datanew)

# # Plot the fitted model

plot1<-ggplot(data=data, aes(x=time_centered))+
  geom_point(data=data, aes(y=rate))+
  geom_line(aes(y=pred1b),
            color="dodgerblue", linetype="dashed")+
  geom_line(aes(y=pred1))+
  expand_limits(y=c(0,300))+
  theme_bw()
```

ITS excersize



Does it look like there is seasonality in this data?

ITS excersize

How can we adjust for seasonality?

- **Time stratification**: Using time fixed effects to model time intervals (months, seasons, years, etc.) (this is how you will do it on the problem set)
- **Periodic functions**: 'Fourier' terms are pairs of sine and cosine functions – assumes seasonality is constant between years
- **Flexible splines**: polynomial curves estimated for discrete intervals, risk of overfitting

ITS excersize

Let's add monthly-fixed effects. Are these effects significant? Does this meaningfully change our estimate of the immediate effect?

```r
model_2 <- lm(rate ~ time_centered + smokban + factor(month), data=data)
summary(model_2)
```

```
##
## Call:
## lm(formula = rate ~ time_centered + smokban + factor(month),
##     data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.1206  -4.3999  -0.3254   5.3983  20.3157
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)      253.1671     5.2339  48.370  < 2e-16 ***
## time_centered      1.1779     0.1363   8.639 4.13e-11 ***
## smokban          -26.9014     4.6977  -5.726 7.94e-07 ***
## factor(month)2   -27.0594     5.6389  -4.799 1.79e-05 ***
## factor(month)3    -2.2884     5.6439  -0.405  0.68706
## factor(month)4   -20.6575     5.6521  -3.655  0.00067 ***
## factor(month)5    -8.9658     5.6636  -1.583  0.12041
## factor(month)6   -29.8605     5.6784  -5.259 3.87e-06 ***
## factor(month)7   -29.8000     5.6963  -5.231 4.24e-06 ***
## factor(month)8   -31.4726     5.7175  -5.505 1.69e-06 ***
## factor(month)9   -31.0667     5.7418  -5.411 2.32e-06 ***
## factor(month)10  -16.5835     5.7693  -2.874  0.00616 **
## factor(month)11  -15.3492     5.7998  -2.646  0.01117 *
## factor(month)12   -4.1428     6.1270  -0.676  0.50241
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

ITS excersize

Instead of hypothesizing that there would only be a level shift in ACEs, you also ypothesize that there is a change in the trend. You estimate the following model:

$$ACE_rate_t = \alpha + \beta time_centered_t + \delta smoke_ban_t + \gamma(time_centered \times smoke_ban)_t + \mu_t + \epsilon_t$$

ITS excersize

How would you interpret these coefficients? Is the coefficient on the time-ban interaction significant?

```r
model_3 <- lm(rate ~ time_centered + smokban + time_centered:smokban + factor(month), data=data)
summary(model_3)
```

```
##
## Call:
## lm(formula = rate ~ time_centered + smokban + time_centered:smokban +
##     factor(month), data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -20.8273  -3.8050  -0.0875   5.7848  19.5715
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)            252.1932     5.2561  47.981  < 2e-16 ***
## time_centered            1.1034     0.1477   7.472 2.34e-09 ***
## smokban                -29.1112     4.9829  -5.842 5.75e-07 ***
## factor(month)2         -27.1456     5.6021  -4.846 1.60e-05 ***
## factor(month)3          -2.4609     5.6083  -0.439 0.662957
## factor(month)4         -20.9163     5.6185  -3.723 0.000557 ***
## factor(month)5          -9.3108     5.6328  -1.653 0.105460
## factor(month)6         -30.2918     5.6512  -5.360 2.91e-06 ***
## factor(month)7         -30.3175     5.6735  -5.344 3.08e-06 ***
## factor(month)8         -32.0764     5.6998  -5.628 1.19e-06 ***
## factor(month)9         -31.7567     5.7300  -5.542 1.58e-06 ***
## factor(month)10        -17.3598     5.7640  -3.012 0.004293 **
## factor(month)11        -16.2117     5.8017  -2.794 0.007676 **
## factor(month)12         -4.2426     6.0871  -0.697 0.489481
## time_centered:smokban    0.4018     0.3176   1.265 0.212528
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

ITS excersize

$$ACE_rate_t = \alpha + \beta time_centered_t + \delta smoke_ban_t + \gamma(time_centered \times smoke_ban)_t + \mu_t + \epsilon_t$$

- β: The average monthly rate of change in the average rate of ACEs per 10,000 people before policy is 1
- δ: Policy implementation immediately reduces the level of ACEs per 10,000 people by 29.
- γ: Policy implementation reduces the monthly rate of change level of ACEs per 10,000 people by 0.4.